Computational Modeling
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Spring term 2026
Project 2, due March 11, 2026 at 11:59 p.m.
to be uploaded to https://elearning.constructor.university

2A. Random walk with Absorbing Boundaries [40 points]

a) Consider a 1D random walker starting at the origin $x = 0$. There are absorbing boundaries (cliffs) located at $x = +L$ and $x = -L$. The walker steps with equal probability $p = q = 0.5$ to the left or right. Numerically simulate the walk and record the time (number of steps) T it takes for the walker to be absorbed (i.e., to reach either $+L$ or $-L$).
b) Compute the Mean First Passage Time $\langle T \rangle$ for different boundary distances $L = 4, 8, 16, 32, 64$. Plot $\langle T \rangle$ as a function of L. Does the time scale linearly with the distance ($T \propto L$) or diffusively ($T \propto L^2$)? Compare your computational results with the analytical prediction for the "Gambler's Ruin" problem.

While the previous problem dealt with constrained simple walkers, the next problem introduces memory effects, modeling polymers rather than gas particles.

2B. Self-Avoiding Walk (SAW) in 2D [60 points]

a) Consider a "polymer" chain modeled as a random walk on a 2D square lattice, starting at the origin. The constraint is that the walk is *self-avoiding*: the walker cannot step on a lattice site that has already been visited during the current walk. Write a program to generate random self-avoiding walks of length N. (Note: If a walker gets trapped and cannot move without intersecting itself, you may discard that attempt and restart, a method known as simple sampling).
b) Generate an ensemble of $M \geq 1000$ successful walks for step counts $N = 5, 10, 15, 20, 25, 30$. (Note: As N increases, the attrition rate of simple sampling increases significantly).
c) Compute the mean square end-to-end distance $\langle R^2 \rangle$ given by:

$$\langle R^2 \rangle = \langle x_N^2 + y_N^2 \rangle$$

where (x_N, y_N) is the position of the N-th segment. How does $\langle R^2 \rangle$ for the Self-Avoiding Walk compare to the simple Random Walk calculated in the previous exercise set?
d) Assume the scaling law:

$$\sqrt{\langle R^2 \rangle} \propto N^\nu$$

Use a log-log plot of $\langle R^2 \rangle$ versus N to determine the scaling exponent ν. Does your result match the theoretical mean-field prediction for a simple random walk ($\nu = 0.5$) or the prediction for polymers ($\nu \approx 0.75$ in 2D)?
e) Calculate the "attrition rate." Plot the fraction of attempted walks that get trapped before reaching step N as a function of N.

The use of Generative AI is not allowed in this course!